Hands-On Lab (HOL-02): Create and Connect to a Windows Virtual Machine in AWS

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1. Introduction: Windows Virtual Machine in AWS

A **Windows Virtual Machine** in AWS is a cloud-based server that runs the **Windows Operating System**.

Instead of installing Windows on a physical laptop or server, AWS allows us to run Windows in the cloud using **EC2**.

You can use a Windows EC2 instance for:

- Windows applications
- IIS web servers
- .NET applications
- Remote Desktop access (RDP)

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2. How Windows EC2 is Different from Linux EC2

Before starting the lab, understand this clearly:

Linux EC2	Windows EC2
Login using SSH	Login using RDP
Uses key pair for SSH	Key pair is used to decrypt password
Username: ec2-user	Username: Administrator

So yes, **key pair is still required** for Windows, but it is used differently.

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3. Lab Objective

By the end of this lab, you will:

- Create a Windows EC2 instance
- Configure instance type, storage, and network
- Retrieve Windows password

- Connect using Remote Desktop (RDP)

Step 1: Login to AWS Console

1. Open browser
2. Go to https://aws.amazon.com
3. Click **Sign In to the Console**
4. Login using your AWS account

Step 2: Open EC2 Service

1. In AWS Console search bar, type **EC2**
2. Click on **EC2**
3. EC2 Dashboard will open

Step 3: Launch a New EC2 Instance

1. Click **Launch Instance**
2. Click **Launch Instance** again

Step 4: Basic Configuration (Name and OS)

4.1 Name the Instance

- Instance Name: `My-Windows-EC2-`

4.2 Choose Amazon Machine Image (AMI)

AMI defines the operating system.

Choose:

- **Microsoft Windows Server 2019 Base** (or 2022)

Why Windows Server?

- Stable
- Widely used in enterprises
- Supports RDP and IIS

Step 5: Choose Instance Type (CPU & RAM)

What is Instance Type?

Instance type defines:

- CPU power
- Memory (RAM)
- Overall performance

Windows OS needs more resources than Linux.

Why are we choosing this?

Choose:

- **t2.micro** (Free Tier eligible, if available)

If t2.micro is not available:

- Choose **t3.micro**

This is enough for:

- Windows login
- Basic configuration
- Learning purpose

Step 6: Key Pair (Very Important for Windows)

Even for Windows, **key pair is mandatory**.

Why key pair is required?

- AWS encrypts the Windows Administrator password
- Key pair is used to **decrypt the password**

Without key pair:

- You cannot retrieve Windows password

Create Key Pair

1. Click **Create new key pair**
2. Key pair name: windows-ec2-
3. Key pair type: RSA

4. Private key format: `.pem`
5. Click **Create key pair**

⚠️ Save the key safely

Step 7: Network Settings (Firewall & Access)

Network settings control how you connect to Windows VM.

7.1 VPC

- Use **Default VPC**

Why?

- Pre-configured
- Internet access enabled
- Best for beginners

7.2 Subnet

- Default subnet

7.3 Auto-assign Public IP

Enable this option.

Why?

- Required to connect using RDP from your laptop

7.4 Security Group (Firewall)

Security group controls incoming traffic.

Create new security group:

- Name: `windows-rdp-`

Allow rule:

- RDP | Port 3389 | Source: My IP

Why port 3389?

- RDP uses port 3389

Why My IP?

- Restricts access
- Improves security

Step 8: Configure Storage

Windows OS needs more storage than Linux.

Storage explanation

- Root volume stores:
- Windows OS
- System files
- Applications

Recommended settings

- Size: **30 GB** (minimum for Windows)
- Volume type: gp2 / gp3

Step 9: Review and Launch

1. Review all settings carefully
2. Click **Launch Instance**
3. Instance will start (Windows takes more time than Linux)

Step 10: Wait for Instance to Become Ready

1. Go to **EC2** → **Instances**
2. Instance state: **Running**
3. Status checks: **2/2 passed**

Note:

- Windows may take 3–5 minutes to be ready

Step 11: Retrieve Windows Administrator Password

1. Select your Windows instance
2. Click **Connect**
3. Choose **RDP Client**
4. Click **Get password**
5. Upload your `.pem` key file
6. Click **Decrypt password**

You will get:

- Username: `Administrat`
- Password

Step 12: Connect to Windows EC2 Using RDP

From Windows Laptop

1. Open **Remote Desktop Connection**
2. Enter Public IP of EC2
3. Click **Connect**
4. Enter:
5. Username: Administrator
6. Password (copied from AWS)

From Mac / Linux

- Install Microsoft Remote Desktop client
- Use Public IP, username, and password

Step 13: Verify Windows VM

After login:

- Desktop should load
- Server Manager will open

You are now inside your Windows EC2 instance ‼️

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4. Clean Up (Mandatory)

To avoid AWS charges:

1. Go to EC2 → Instances
2. Select Windows instance
3. Click **Instance state** → **Terminate instance**

Lab Summary

In this lab, you learned:

- How to create Windows EC2 instance
- Difference between Linux and Windows EC2
- How key pair is used for Windows password
- How to connect using RDP

✅ End of Hands-On Lab (HOL-02)